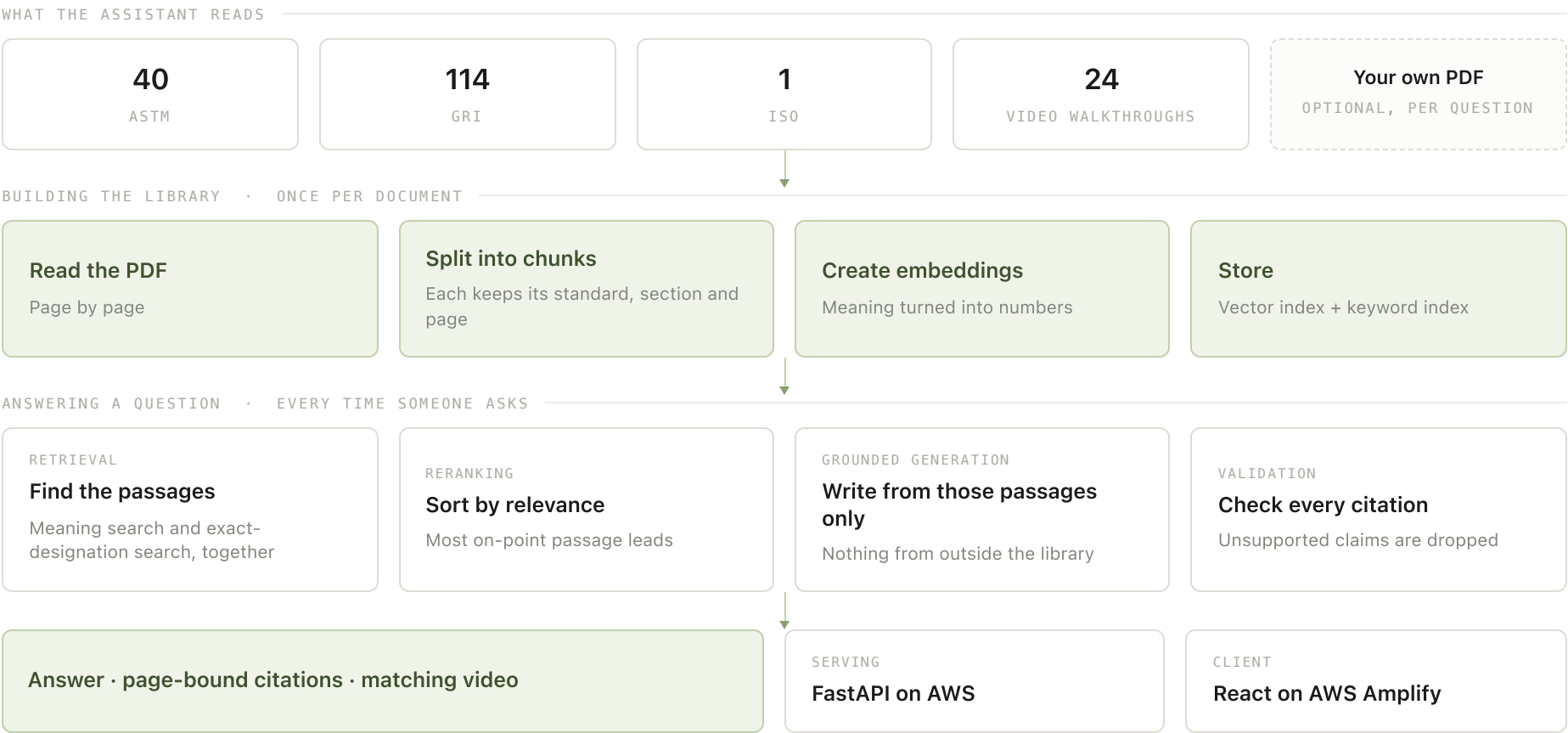


FIGURE 1 System architecture



IN PLAIN LANGUAGE

Chunk - a short passage of a standard.

Keyword index - catches exact codes like D5199.

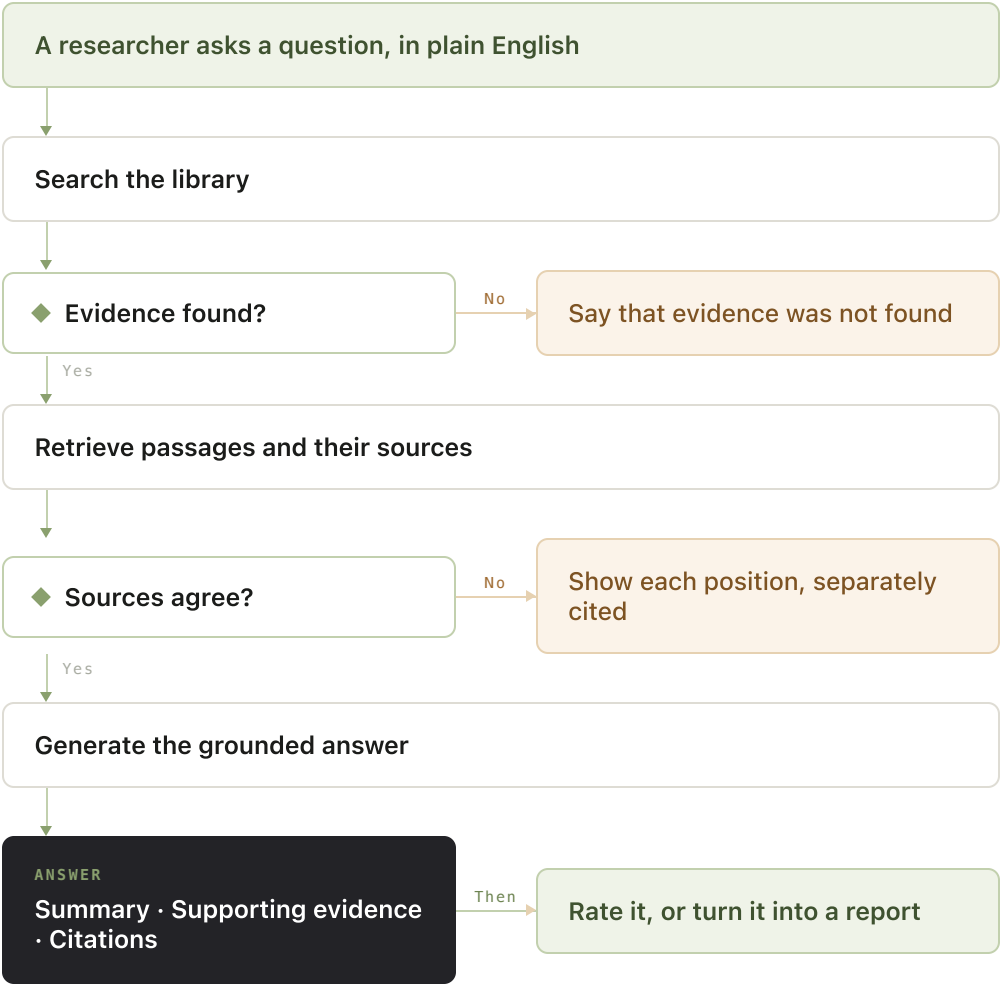
Embedding - a numeric fingerprint of meaning.

Reranking - puts the best passage first.

Vector index - finds meaning, not exact words.

Grounded - written only from what was found.

FIGURE 2 User flow



After the answer

Helpful LIVE

Not helpful LIVE
Opens a comment box

Generate report → PDF NEXT
Selected answers exported as one document

THE REPORT CARRIES

Question · answer · every citation with its page · link to each source · date

BACK TO THE LIBRARY

A disliked answer usually means a missing standard. Ratings land in the admin portal, next to the question that produced them.

FIGURE 3 Asking across the literature

GSI Chatbot

+ New chat

Search chats

Admin portal

ChatsProjects

SAVED RESEARCH THREADS

D5199 Thickness Measurement

2 messages

D5199 Thickness Measurement

2 messages

D5397 Stress Cracking

2 messages

member@gsi.org

Sign out

GSI Chatbot

Standards Q&A — grounded in your ASTM, ISO, and GRI index

1

What thickness do the standards require for an HDPE geomembrane liner, and do the different standards makers agree with each other?

The standards for HDPE geomembrane liners do not specify a uniform thickness requirement, but they do provide guidelines on how to measure and assess the thickness of these materials.

- Thickness Measurement:** According to ASTM D5199, the thickness measurement instruments must be precise, with a base and a free-moving presser foot plate that are parallel to each other within a tolerance of less than 0.01 mm. This ensures accurate thickness readings of the geomembrane liners [1].
- Referenced Standards:** The GRI GM37 standard also references ASTM D5199 among other ASTM standards, indicating a consensus on the importance of these measurement practices for geomembranes. This suggests that the methodologies for assessing thickness are recognized across different standards [2].
- Test Specimen Selection:** Furthermore, ASTM D5397 emphasizes the importance of selecting appropriate test specimens, which can include the geomembrane as received, molded plaques, or smooth edges. This flexibility allows for various testing conditions to be considered when determining thickness [3].

In summary, while the standards do not mandate a specific thickness for HDPE geomembranes, they agree on the importance of precise measurement techniques and the selection of appropriate specimens for testing. This consensus among different standards bodies indicates a shared understanding of the critical aspects of geomembrane thickness evaluation.

CITATIONS

- [D5199-25, Standard Test Methods for Measuring the Nominal Thickness of Geosynthetics](#)¹ This standard is issued under the fixed designation D5199; the number immediately following the designation indicates the year of, Section 5.3, page 2
- [GRI-GM37r, Standard Test Method for "Determining the Weld Edge Thickness of Geomembrane"](#), page 1
- [D5397-20, Standard Test Method for Evaluation of Stress Crack Resistance of Polyolefin Geomembranes Using Notched Constant Tensile Load Test](#)¹, Section 4, page 6

4

+ Ask anything about standards...

>

GRI

GEI

GSI

GAI

GCI

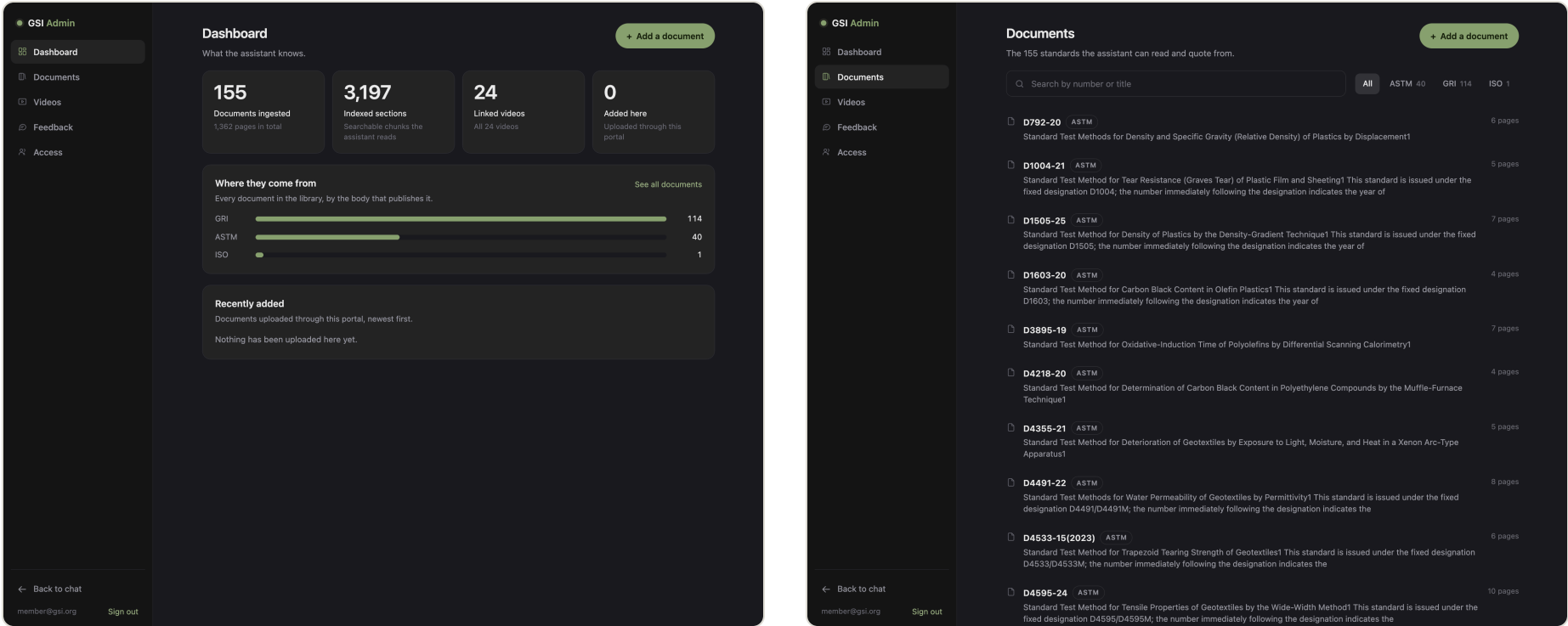
1 A question no single standard answers

2 Inline markers tie each claim to a passage

3 Section and page on every citation

4 Three publishing bodies, cited separately

FIGURE 4 The administrator's portal



Dashboard. What the assistant knows, and where it came from.

Documents. The full library, filterable by publishing body.

ADDING A STANDARD

STEP 1

Drop in the PDF

The portal proposes number, title, body, year

STEP 2

Confirm or correct

Nothing is written until this point

ON CONFIRM

Answerable immediately

Re-uploading replaces, never duplicates

ALSO IN THE PORTAL

Videos · Feedback · Access

Walkthroughs, rated answers, who may curate

FIGURE 5 Extending the platform to IGS

